

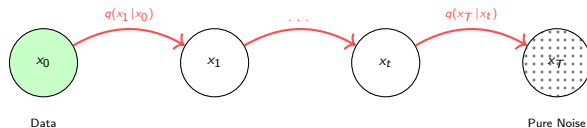
Scaling Rectified Flow Transformers for High-Resolution Image Synthesis

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Fundamental Team

November 23, 2025

1. Preliminary: DDPM Forward Process (Diffusion)



The Forward Process (Structure Destruction)

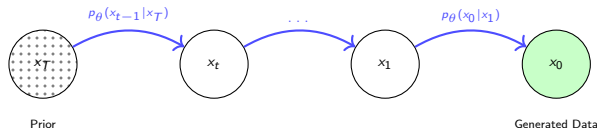
1. Step-wise Transition: A Markov chain that gradually adds Gaussian noise according to a variance schedule β_t .

$$q(x_t|x_{t-1}) := \mathcal{N}(x_t; \sqrt{1 - \beta_t}x_{t-1}, \beta_t \mathbf{I})$$

2. The "Nice Property" (Closed Form): We can sample x_t directly from x_0 without iterating steps. Let $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$.

$$q(x_t|x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_0, (1 - \bar{\alpha}_t)\mathbf{I}) \implies \boxed{x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon}, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I})$$

2. Preliminary: DDPM Reverse Process (Denoising)



The Reverse Process (Structure Restoration)

1. The Generative Model: Since $q(x_{t-1}|x_t)$ is intractable, we approximate it with a learnable model p_θ .

$$p_\theta(x_{t-1}|x_t) := \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t))$$

2. Training Objective (Simplified MSE): Instead of predicting the mean μ_θ directly, Ho et al. (2020) found it better to predict the noise ϵ .

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{t, x_0, \epsilon} \left[\|\epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon, t)\|^2 \right]$$

By minimizing the difference between actual noise (ϵ) and predicted noise (ϵ_θ)

3. Preliminary: The Continuous Limit (SDE)

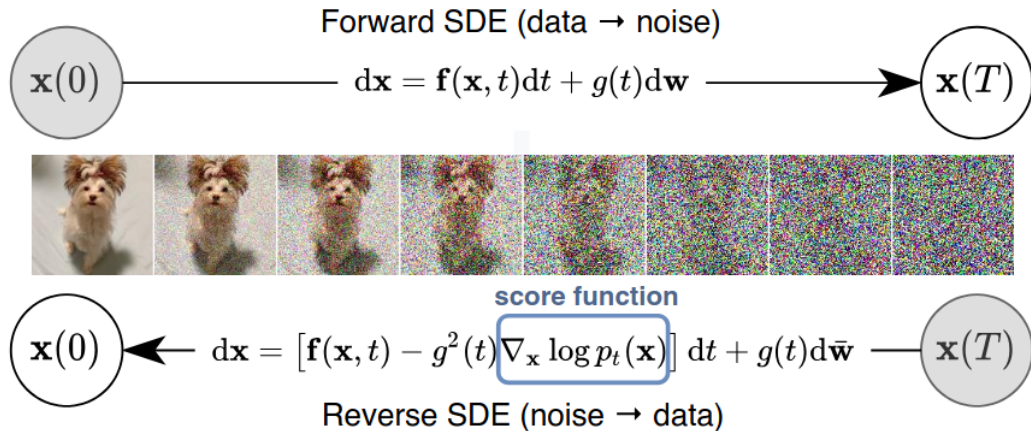


Figure: Solving a reverse-time SDE yields a score-based generative model

3. Preliminary: The Continuous Limit (SDE) - Overview

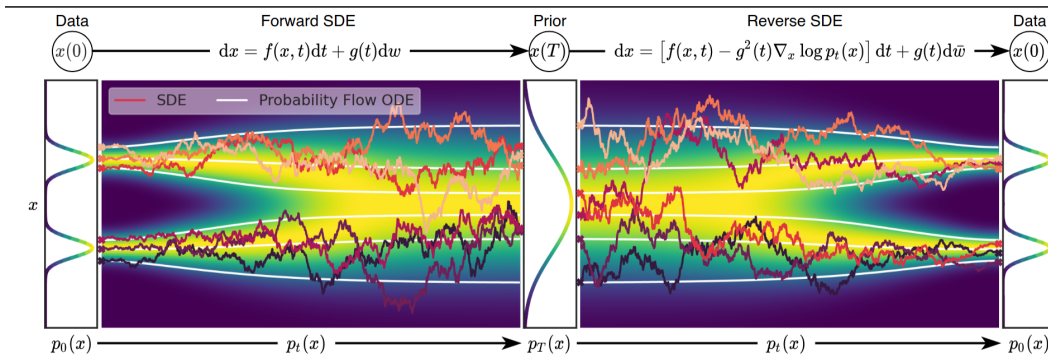


Figure: Overview of score-based generative modeling through SDEs

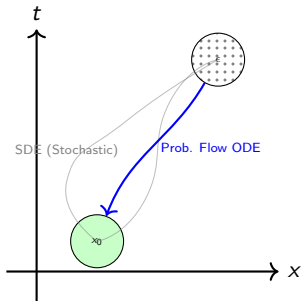
3.1. Preliminary: From Stochastic (SDE) to Deterministic (ODE)

The Probability Flow ODE (Song et al., 2020)

For any SDE that diffuses data to noise, there exists an **Ordinary Differential Equation (ODE)** that shares the exact same marginal probability densities $p_t(x)$.

Why does this matter?

- **Deterministic Sampling:** We can turn noise into images without random injection.
- **Invertibility:** We can encode images to noise and back.



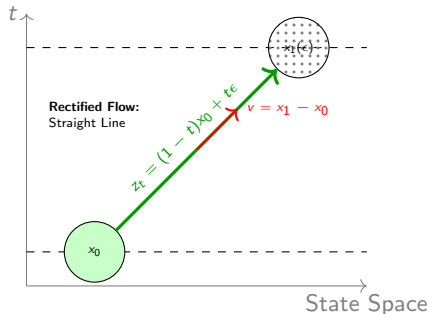
The Problem with Standard Diffusion ODEs

The trajectory is **curved** and complex.

$$dx = \left[f(x, t) - \frac{1}{2} g(t)^2 \nabla_x \log p_t(x) \right] dt$$

Solving this requires many steps (expensive inference).

3.2. Preliminary: The SD3 Approach: Rectified Flows (Straight ODE)



Defining a Straight Path

Instead of deriving the ODE from an SDE, we **define** the forward process as a linear interpolation:

$$z_t = (1 - t)x_0 + t\epsilon$$

- $t = 0$: Data (x_0)
- $t = 1$: Noise (ϵ)

The Velocity Matching Objective

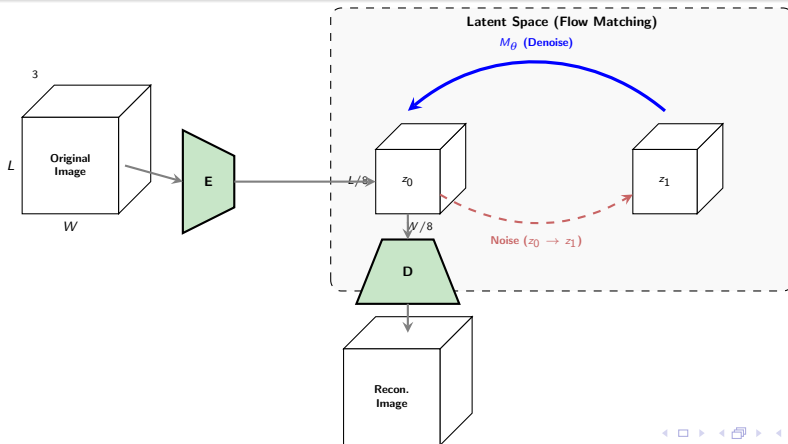
The velocity $v_t = \frac{dz_t}{dt}$ is simply $x_1 - x_0$. The model v_θ learns to predict this straight direction:

$$\mathcal{L}_{FM} = \mathbb{E}_{t, x_0, \epsilon} \left[\left\| v_\theta(z_t, t) - \underbrace{(\epsilon - x_0)}_{\text{Target Velocity}} \right\|^2 \right]$$

4. Preliminary: Connection to Latent Space

Where does the Flow happen?

- Running diffusion/flow on high-res pixels ($1024 \times 1024 \times 3$) is computationally prohibitive.
- Solution:** Use a VAE (Variational Autoencoder) to compress images into a lower-dimensional **Latent Space**.



1. Introduction: The Efficiency Bottleneck

- **The Current Landscape:**

- Diffusion models are the de-facto standard for high-resolution image and video generation.
- They generate data by inverting a forward path from data to noise.

- **The Problem:**

- **Iterative Nature:** High computational cost and slow sampling times during inference.
- **Curved Paths:** Standard diffusion models often learn "curved" trajectories from noise to data, requiring many integration steps.

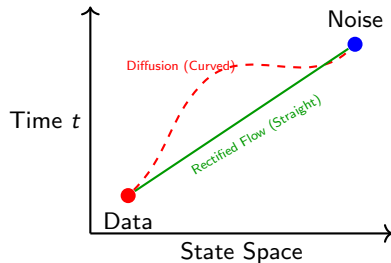
2. The Solution: Rectified Flows (RF)

- **Rectified Flow Concept:**

- A generative formulation that connects data (x_0) and noise (x_1) in a **straight line**.
- *Theory*: A straight path can theoretically be simulated in a single step, reducing error accumulation.

- **The Challenge:**

- Despite better theoretical properties, RF has not yet established itself as standard practice.
- This paper aims to fix this by improving noise sampling techniques.



3. Architectural Limitations: Text-to-Image

- **Standard Architecture (e.g. Stable Diffusion 1.5/2):**
 - Uses a fixed text representation fed directly into the model via **Cross-Attention**.
 - *Paper Argument:* This is "not ideal" as it restricts the flow of information between modalities.
- **The Proposed Shift:**
 - Need for an architecture that allows **learnable streams** for both image and text tokens.
 - Enables bidirectional information flow (image affects text representation, text affects image).

4. Key Contributions

1. Improved Rectified Flow Formulation

- Introduced new **noise sampling** techniques (Logit-Normal Sampling) biased towards perceptually relevant scales.
- Outperforms established diffusion formulations for high-res synthesis.

2. MM-DiT Architecture

- A novel **Multimodal Diffusion Transformer** .
- Uses separate weights for text and image modalities but mixes them via joint attention.

3. Scaling Laws & Performance

- Demonstrated that validation loss predicts downstream performance (CompBench, GenEval, Human Eval).
- Scaling up to **8B parameters** outperforms state-of-the-art (SDXL, DALL-E 3).

1. The Mathematical Foundation: Rectified Flows

Core Concept: Connecting data (x_0) and noise (x_1) via a straight line.

The forward process is defined as:

$$z_t = (1 - t)x_0 + t\epsilon, \quad \text{where } \epsilon \sim \mathcal{N}(0, I)$$

The Learning Objective: The model v_θ learns to predict the velocity (direction) to move from noise back to data.

Flow Matching Objective

$$\mathcal{L} = \mathbb{E}_{t, \epsilon, x_0} \left[w_t \|v_\theta(z_t, t, c) - (x_1 - x_0)\|^2 \right]$$

- z_t : The noisy input at time t .
- t : The timestep (noise level).
- c : The text conditioning (prompt).
- v_θ : The Neural Network (MM-DiT) output.

1. Training: Linear Interpolation

Unlike curved diffusion, we use a simple **linear mix** of Data and Noise.

Forward Equation (x_t):

$$x_t = (1 - t)x_0 + t\epsilon$$

- $t = 0 \rightarrow$ Data (x_0)

- $t = 1 \rightarrow$ Noise (ϵ)

Loss Function: $\mathcal{L} = \text{MSE}(v_\theta(x_t), \epsilon - x_0)$ (Model learns the "slope")

2. Inference: Reconstruction Algebra

How do we create an image? (Reversing the linear process)

Step A: Predict Original (x_0)

$$\hat{x}_0 = \frac{x_t - t\hat{\epsilon}}{1 - t}$$

Step B: Move to Next ($x_{\hat{t}}$)

$$x_{\hat{t}} = (1 - \hat{t})\hat{x}_0 + \hat{t}\hat{\epsilon}$$

**This corresponds to the Euler Step visualized in the notes.*

2. MM-DiT Backbone Analysis (1/2): Inputs & Modulation

1. Detailed Input Processing

- **Text Encoders:**

- CLIP-G/14, CLIP-L/14, T5 XXL used.

- **Sequence (c_{ctx}):**

- Concat. of outputs (77+77 tokens).
- Projected to 4096 channels.

- **Vector (y):**

- Pooled CLIP-G + Timestep $\rightarrow$ MLP.

- **Image (x):**

- VAE Latent $\rightarrow 2 \times 2$ Patching.

2. MM-DiT Block Structure

- **Dual Stream:**

- Separate weights for text & image.
- Preserves distinct modality features.

- **Joint Attention:**

- *Concat* $\rightarrow$ *Attention* $\rightarrow$ *Split*.
- Enables bidirectional flow.
- Q, K, V computed jointly.

- **Outcome:**

- Improved typography & prompt following.

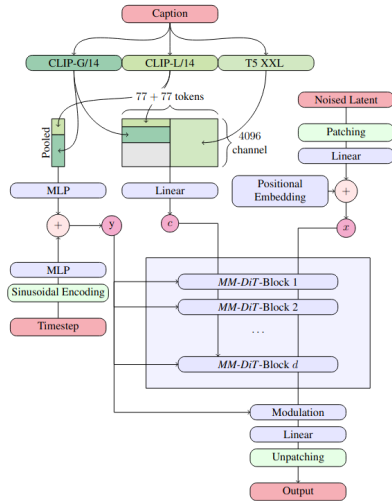
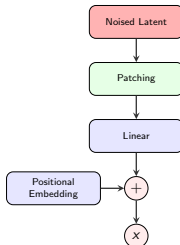
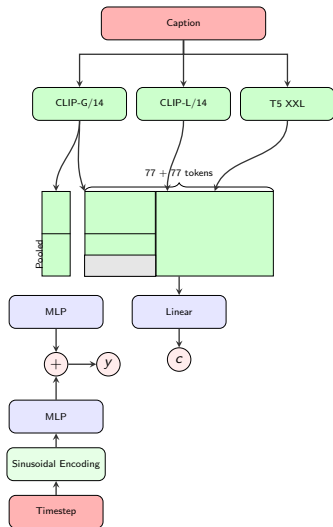


Figure: Overview of all components

1. Input Processing Overview (The Big Picture)

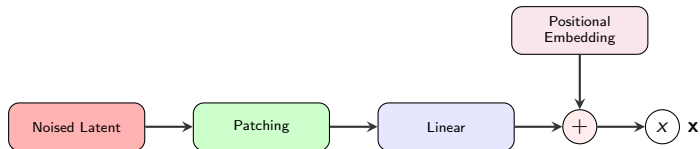


Three Key Input Streams:

- 1 **Modulation (y)**
Controls global style (Timestep + Pooled Text).
- 2 **Text Context (c)**
Detailed semantic tokens (CLIP + T5 concat).
- 3 **Image Latent (x)**
The actual data to be denoised.

Let's break these down one by one →

2. Breakdown: Image Stream ($\mathbf{x}$)



1. VAE Latent

- Input: z_t
- Compressed visual data with noise.

2. Patching

- 2×2 patches.
- Flattened to 1D sequence.

3. Linear Proj.

- Maps to model dimension D .
- $\text{Linear}(\text{Patch}(z_t))$

4. Pos. Embedding

- Injects 2D spatial info.
- $\mathbf{x} = \text{Proj} + \mathbf{P}_{img}$

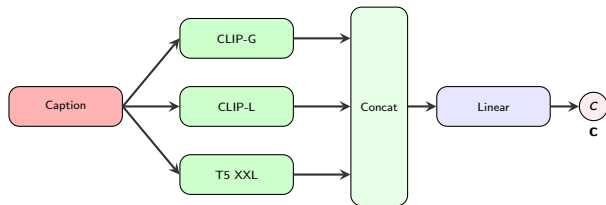
3. Breakdown: Text Context Stream (c)

Why Multi-Encoder?

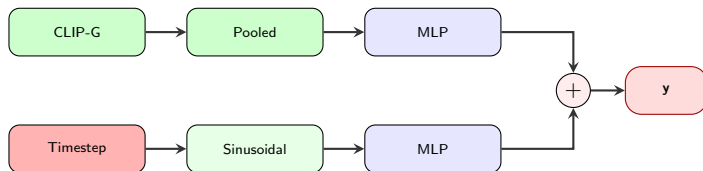
- **CLIP**: Visual alignment.
- **T5**: Complex semantic understanding.

Integration:

- Outputs are **concatenated** into a single long sequence.
- Mapped to dimension D via Linear.



4. Breakdown: Modulation Vector (y)



1. Timestep Signal

- Input: Current noise level t .
- Encoding: Sinusoidal (Freq).
- Output: $\text{MLP}(\text{Sin}(t))$

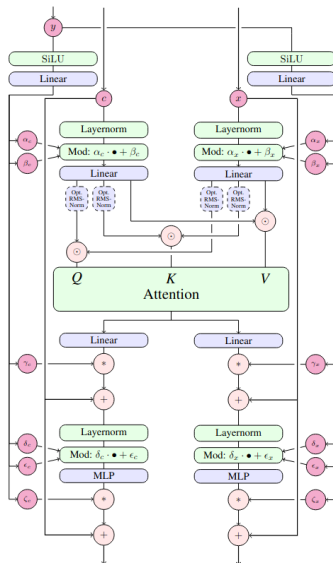
2. Global Text Context

- Input: CLIP-G [EOS] token.
- Pooling: Summary of caption.
- Output: $\text{MLP}(\text{Pooled})$

3. Fusion & Control

- Operation: Element-wise Sum (+).
- Role: y regresses γ, β for AdaLN.
- Controls scale/shift per block.

MM-DiT Block: The Big Picture



Breakdown 1: Modulation (Adaptive LayerNorm)

Signal Conditioning

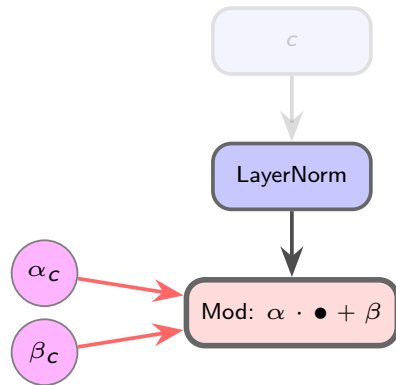
Conditioned on timestep t and vector y . Regressed into scale (α) and shift (β).

Equation:

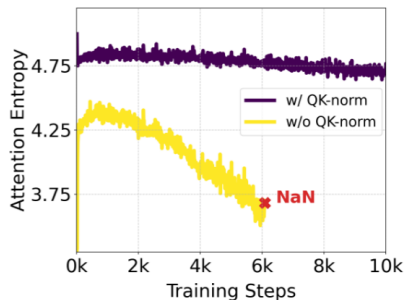
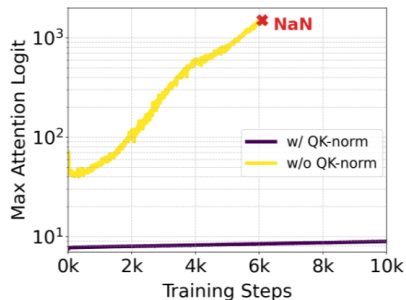
$$\text{Mod}(u) = (1 + \alpha_x) \odot u + \beta_x$$

Parameters:

- α, β : Pre-Attention
- δ, ϵ : Pre-MLP
- γ, ζ : Gating



Breakdown 2: QK-Norm & Projections



The Problem: Instability

Scaling to large models (e.g., 8B) causes attention logits to grow uncontrollably (**entropy collapse**).

The Solution: QK-Norm

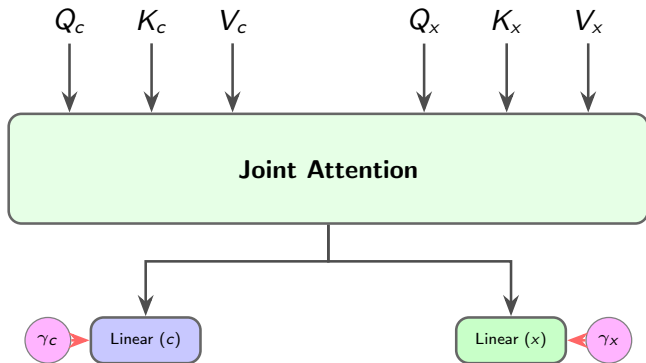
Apply **RMSNorm** to Query (Q) and Key (K) vectors *before* attention.

Breakdown 3: Joint Attention Mechanism

Sequence Concatenation:

$$Z = \text{Concat}([Q_c; Q_x], \dots)$$

Split & Project: Output split back into separate streams.



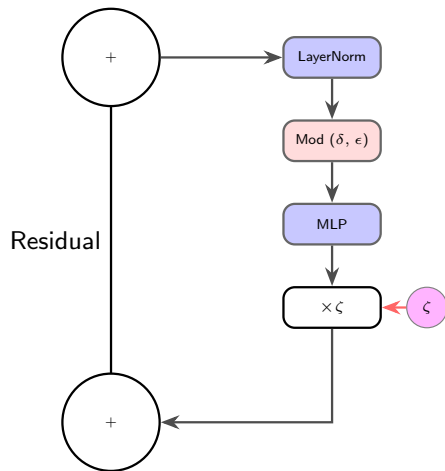
Breakdown 4: MLP & Residual Connections

Feed-Forward

Pointwise MLP independently applied to each stream.

Equation:

$$x_{next} = x_{prev} + \zeta \cdot \text{MLP}(\text{Mod}(x))$$



4. Mapping the Formula to the Architecture

The paper proposes the **MM-DiT** to solve the velocity prediction $v_\theta(z_t, t, c)$.

The Input Variables

① Noisy Image (z_t):

- Enters the **Image Stream**.
- Processed as flattened patches.

② Text Condition (c):

- Enters the **Text Stream**.
- Uses separate weights (Transformers) to preserve text structure.

The Process (t)

- **Timestep** (t): Injected via **AdaLN** (Adaptive Layer Norm).
- It tells the network "how much noise" is currently in the image, scaling the weights dynamically.

The Output (v_θ)

- The final output of the Image Stream is the predicted velocity vector (the denoising direction).

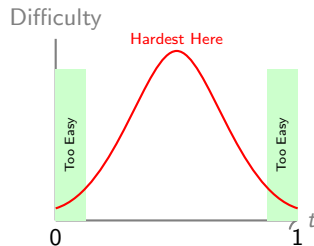
1. The Problem: Not All Timesteps Are Equal

Standard Rectified Flow (RF)

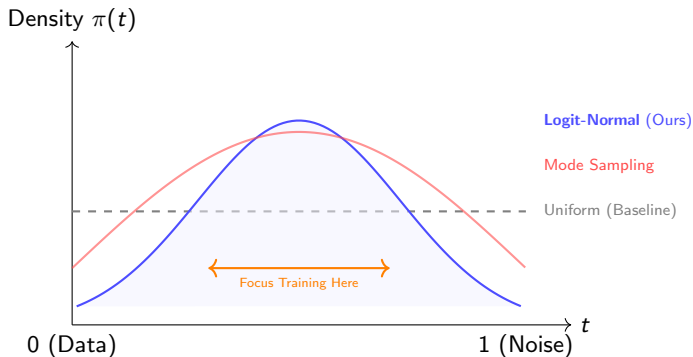
- Conventionally samples time t uniformly: $t \sim \mathcal{U}(0, 1)$.
- **Issue:**
 - $t = 0$ (Data): Trivial (Identity).
 - $t = 1$ (Noise): Trivial (Mean prediction).
 - The **Middle** is where the complex structural formation happens.

Hypothesis

Training should be biased towards **intermediate timesteps** where the velocity v_t is most difficult to predict.



2. The Solution: Biasing the Timestep Schedule



Logit-Normal Sampling (π_{ln}):

$$\pi_{ln}(t; m, s) = \frac{1}{s\sqrt{2\pi t(1-t)}} e^{-\frac{(\text{logit}(t)-m)^2}{2s^2}}$$

- **Bell-shaped** on $(0, 1)$.

Key Findings:

- Assigns **more weight** to the middle.
- Suppresses learning at trivial endpoints.
- **Winner:** $m = 0.00, s = 1.00$.

3. Experimental Results (Ablation)

The authors tested 61 different formulations. The **Logit-Normal** schedule significantly outperformed existing methods.

Summary Data:

Variant	Rank	Note
rf/lognorm	1.54	Ours
rf/lognorm (bias)	2.08	Too biased
eps/linear	2.88	SD 1.5/2.1
rf/cosmap	4.13	
edm	5.63	
rf (uniform)	5.67	Baseline

Conclusion

Changing the **timesteps distribution** (t) to focus on the "difficult" middle section yields better performance than changing the vector field formulation itself.

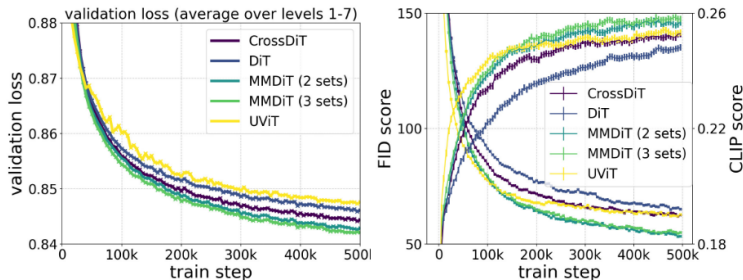
variant	rank averaged over		
	all	5 steps	50 steps
rf/lognorm(0.00, 1.00)	1.54	1.25	1.50
rf/lognorm(1.00, 0.60)	2.08	3.50	2.00
rf/lognorm(0.50, 0.60)	2.71	8.50	1.00
rf/mode(1.29)	2.75	3.25	3.00
rf/lognorm(0.50, 1.00)	2.83	1.50	2.50
eps/linear	2.88	4.25	2.75
rf/mode(1.75)	3.33	2.75	2.75
rf/cosmap	4.13	3.75	4.00
edm(0.00, 0.60)	5.63	13.25	3.25
rf	5.67	6.50	5.75
v/linear	6.83	5.75	7.75
edm(0.60, 1.20)	9.00	13.00	9.00
v/cos	9.17	12.25	8.75
edm/vcos	11.04	14.25	11.25
edm/rf	13.04	15.25	13.25
edm(-1.20, 1.20)	15.58	20.25	15.00

Figure A: Global Ranking of Variants (Lower is better)

variant	ImageNet		CC12M	
	CLIP	FID	CLIP	FID
rf	0.247	49.70	0.217	94.90
edm(-1.20, 1.20)	0.236	63.12	0.200	116.60
eps/linear	0.245	48.42	0.222	90.34
v/cos	0.244	50.74	0.209	97.87
v/linear	0.246	51.68	0.217	100.76
rf/lognorm(0.50, 0.60)	0.256	80.41	<u>0.233</u>	120.84
rf/mode(1.75)	0.253	44.39	0.218	94.06
rf/lognorm(1.00, 0.60)	0.254	114.26	0.234	147.69
rf/lognorm(-0.50, 1.00)	0.248	<u>45.64</u>	0.219	89.70
rf/lognorm(0.00, 1.00)	0.250	45.78	0.224	<u>89.91</u>

Figure B: Detailed FID and CLIP Scores

4.1. Architecture Comparison: Why MM-DiT?



Training Dynamics

- **MM-DiT (Green)** achieves lower validation loss faster than standard DiT (Blue) or UViT (Yellow).
- Separate weights effectively handle the conflict between text and image modalities.

Performance Gap

MM-DiT outperforms Cross-Attention baselines (CrossDiT) in both FID and CLIP scores, validating the "Joint Attention" approach.

4.2. Efficiency: Robustness to Few-Step Sampling

Rectified Flows Advantage

Straight paths are easier to simulate, allowing for high-quality generation even with few steps.

Observation (FID Score):

- **RF (Green lines)** degrade gracefully as sampling steps decrease (50 $\rightarrow$ 10).
- Standard Diffusion (Purple/Pink) suffers severe quality drops at low steps.
- **Winner:** rf/lognorm stays competitive even at very low steps.

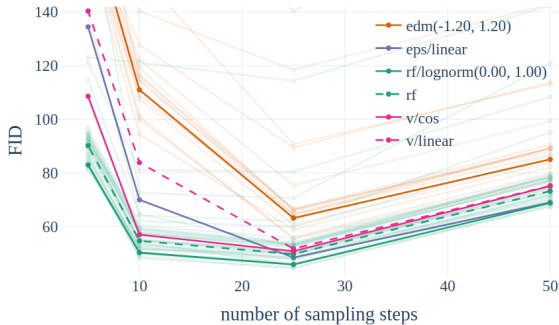


Figure: FID scores across sampling steps (Lower is better)

4.3. Component Improvements: VAE & Captions

1. Improved Autoencoder (VAE)

Metric	4 chn	8 chn	16 chn
FID ($\downarrow$)	2.41	1.56	1.06
Perceptual Similarity ($\downarrow$)	0.85	0.68	0.45
SSIM ($\uparrow$)	0.75	0.79	0.86
PSNR ($\uparrow$)	25.12	26.40	28.62

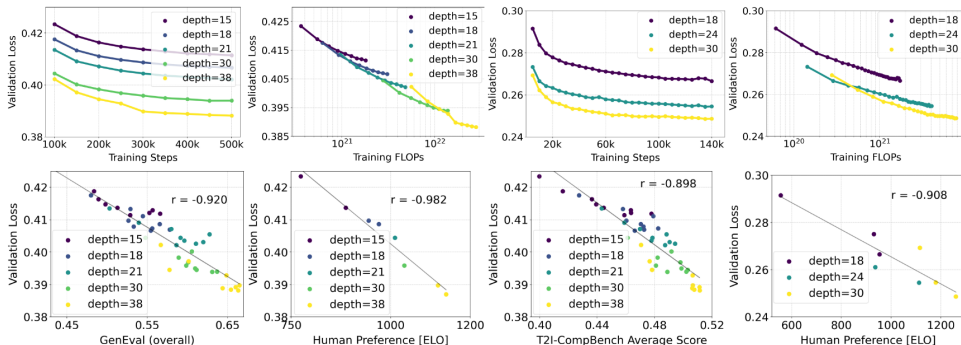
- Increasing latent channels ($4 \rightarrow 16$) drastically reduces reconstruction error (FID $2.41 \rightarrow 1.06$).
- Chosen:** 16-channel VAE for better detail preservation.

2. Synthetic Captions

	Original Captions	50/50 Mix
	success rate [%]	success rate [%]
Color Attribution	11.75	24.75
Colors	71.54	68.09
Position	6.50	18.00
Counting	33.44	41.56
Single Object	95.00	93.75
Two Objects	41.41	52.53
Overall score	43.27	49.78

- Human captions often miss details.
- 50/50 Mix** (Original + Synthetic) outperforms using original captions alone, especially in "Color Attribution" and "Counting".

4.4. Scaling Laws: Predictable Performance



Findings:

- 1 **Power Law:** Validation loss decreases smoothly as model size (depth) and compute (FLOPs) increase.
- 2 **Strong Correlation:** Lower validation loss directly predicts downstream metrics (GenEval Scores, Human Preference).

Implication

We can reliably predict downstream performance just by monitoring validation loss.

4.5. High-Resolution Adaptation: Timestep Shifting

The Issue: High-resolution images have more pixels (signal), requiring "more noise" to fully destroy information. Standard schedules fail here.

The Fix: Shifting (t_m) We shift the timestep schedule towards the noise end.

$$t_m = \frac{\sqrt{m/nt_n}}{1 + (\sqrt{m/n} - 1)t_n}$$

**Top Right graph shows human preference peaking at shift factor 3.0 for 1024px resolution.*

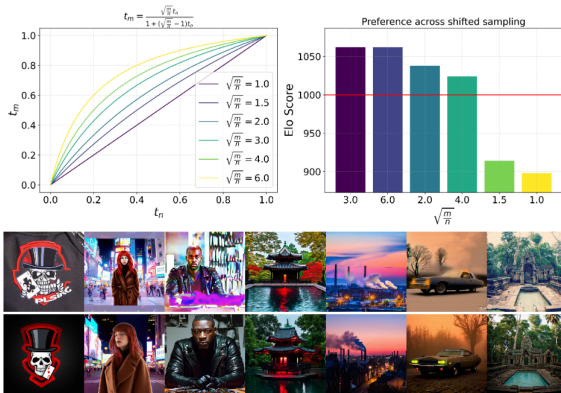
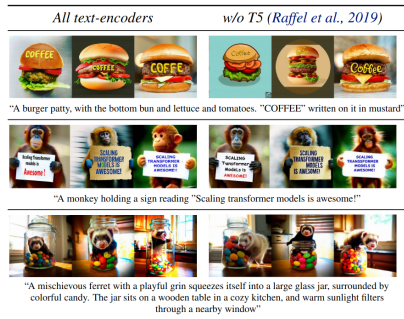


Figure: Effect of Timestep Shifting on Image Quality

4.6. The Role of Text Encoders (T5)



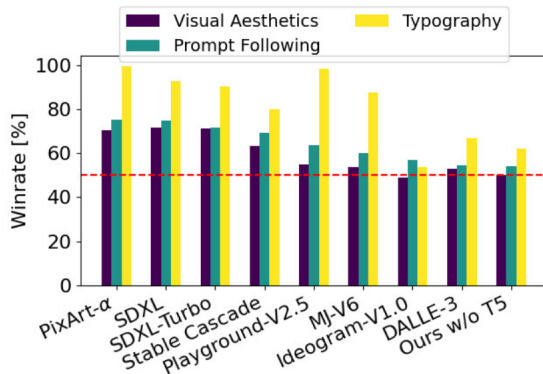
With T5 (Left):

- Precise typography ("COFFEE", "Awesome!").
- Follows complex, detail-heavy prompts.

Without T5 (Right):

- Visual aesthetic remains high (CLIP is sufficient).
- Spelling and dense semantic details degrade.
- *Trade-off: T5 can be removed during inference to save VRAM if text is not required.*

4.7. Final Evaluation: Human Preference



State-of-the-Art Performance

Our 8B model (Rightmost) achieves competitive or superior win-rates against DALL-E 3 and Midjourney v6 in **Typography** and **Prompt Following**.

5. Key Contributions

① Improved Noise Sampling:

- Moved away from uniform sampling to **Logit-Normal Sampling**.
- Biases training towards "middle" timesteps where the task is hardest.

② MM-DiT Architecture:

- Separate weights for text and image modalities.
- Allows bidirectional information flow via joint attention.
- Outperforms UViT and standard DiT backbones.

③ Scaling Laws:

- Validated that lower validation loss correlates with higher human preference.
- Demonstrated performance gains up to 8B parameters.